Copy of AP Calc Unit 1 Test

Answer Key

1. C
2. D
3. A
4. D
5. B
6. B
7. C
8. C
9. B
10. A
11. C
12. B

13. a. $x = 2$

b. $c = 12$

c. Yes; $\lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} (3x + 4) = 3(-2) + 4 = -2$

$\lim_{x \to -2^+} f(x) = \lim_{x \to -2^+} h(x) = -2$

Since $\lim_{x \to -2} f(x) = f(-2) = -2$, f is continuous at $x = -2$.

d. $x = 10$; The graph of f is linear and thus continuous for $x < -2$ and continuous again for $4 \leq x \leq 8$ since g is continuous. When a graph is continuous it cannot have any discontinuities. For $-2 \leq x < 4$, the graph of f is given by $y = h(x)$ which has no infinite discontinuities. For $x > 8$, f has discontinuities when $2(x - 10)(x + 4) = 0 \implies x = 10, \ x = -4$, but only $x = 10$ is in the domain. Since $x = 10$ is not a zero of the numerator, we can conclude that $x = 10$ is an infinite discontinuity for $x > 8$. Therefore, the only infinite discontinuity for the graph of f is at $x = 10$.

e. There is a minimum of 4 solutions to $f(x) = 0$ for $-2 \leq x \leq 8$ since there are three solutions on $-2 \leq x < 4$ and we are guaranteed at least one more solution on the interval $4 \leq x \leq 8$ by the Intermediate Value Theorem. Since g is a continuous function, it must attain a value of 0 on the interval $4 \leq x \leq 8$ since $g(4) > 0$ and $g(8) < 0$.

14. a. $\dfrac{f(5.001) - f(4.999)}{5.001 - 4.999}$ or $\dfrac{7.878 - 8.122}{5.001 - 4.999}$

b. A removable discontinuity; $\lim_{x \to 2} f(x) = 3$ but $f(2) = 6$.

c. $\lim_{x \to 2} 3f(x) - \lim_{x \to 5} \dfrac{f(x)}{2} = 3 \lim_{x \to 2} f(x) - \dfrac{1}{2} \lim_{x \to 5} f(x) = 3(3) - \dfrac{8}{2} = 5$